

Native LaECE (%) with 95 % image-bootstrap CI

RT-DETR-l / D-Fire	13.8 [12.9, 14.6]	16.0 [15.0, 17.1]	26.8 [23.8, 29.9]	40.6 [36.6, 44.1]
RT-DETR-l / Pyro-SDIS	24.4 [21.9, 27.5]	26.2 [21.7, 31.6]	7.3 [6.2, 8.9]	51.9† [48.2, 55.9]
RT-DETR-x / D-Fire	14.3 [13.5, 15.1]	16.6 [15.6, 17.7]	24.9 [22.3, 27.4]	31.5 [27.5, 35.6]
RT-DETR-x / Pyro-SDIS	11.1 [8.5, 16.2]	10.9† [8.1, 19.5]	4.1 [3.2, 5.8]	43.9† [40.1, 48.7]
YOLOv8-m / D-Fire	3.8 [3.3, 4.8]	6.0 [5.1, 7.2]	11.0 [8.9, 14.2]	18.0 [15.7, 22.2]
YOLOv8-m / Pyro-SDIS	22.3 [18.8, 26.2]	25.3† [18.8, 34.3]	2.8 [2.4, 4.7]	54.2† [41.8, 62.1]
YOLOv8-n / D-Fire	2.8 [2.7, 3.9]	3.4 [3.2, 4.7]	7.2 [5.1, 11.0]	17.8 [15.3, 22.6]
YOLOv8-n / Pyro-SDIS	24.1 [20.0, 28.2]	23.8† [17.2, 32.0]	2.4 [2.3, 4.6]	58.9† [46.8, 68.9]
YOLOv8-s / D-Fire	2.6 [2.4, 3.6]	3.8 [3.4, 5.2]	5.6 [4.4, 9.9]	21.3 [18.1, 25.6]
YOLOv8-s / D-Fire de-duplicated	7.6 [6.7, 8.5]	6.4 [5.7, 7.7]	19.9 [16.0, 24.0]	22.1 [18.6, 27.7]
YOLOv8-s / D-Fire leaky subset	2.4 [2.3, 3.6]	3.1 [3.1, 4.7]	9.4 [7.2, 13.1]	18.0 [14.9, 23.0]
YOLOv8-s / Pyro-SDIS	20.2 [16.4, 24.3]	23.7† [18.1, 30.1]	2.0 [2.1, 4.1]	44.8† [33.3, 54.1]
YOLO11-m / D-Fire	2.7 [2.4, 3.8]	4.0 [3.5, 5.4]	10.5 [8.2, 13.9]	14.9 [12.8, 19.7]
YOLO11-m / Pyro-SDIS	21.1 [17.7, 24.9]	19.3 [14.1, 26.0]	2.3 [2.1, 4.2]	47.5† [36.5, 61.5]
YOLO11-n / D-Fire	3.0 [2.8, 4.0]	3.6 [3.4, 5.1]	12.7 [9.6, 17.0]	16.4 [14.0, 21.0]
YOLO11-n / Pyro-SDIS	22.3 [18.5, 26.2]	23.5† [18.2, 31.8]	3.7 [2.9, 5.4]	44.3† [32.4, 52.9]
YOLO11-s / D-Fire	3.5 [3.3, 4.5]	4.4 [3.9, 5.8]	12.5 [10.0, 16.0]	16.8 [14.3, 20.8]
YOLO11-s / D-Fire de-duplicated	4.6 [3.9, 5.6]	3.7 [3.4, 5.2]	16.3 [13.5, 20.3]	21.0 [17.5, 25.4]
YOLO11-s / Pyro-SDIS	20.4 [17.2, 24.7]	21.0† [16.3, 28.9]	2.6 [2.4, 4.6]	39.5† [24.8, 53.3]

D-Fire test D-Fire clean Pyro-SDIS Thesis set

† LRP-FN > 0.95: the model detects almost nothing in this domain, so LaECE is largely the confidence of false positives